Parent Guide — primary / module-12-movement-of-sun

IKS Curriculum

Table of Contents

Parent Guide — Movement of the Sun (Primary)

What is your child learning?

Earth's axial tilt produces an apparent annual path of the Sun along the ecliptic. Indian astronomy names the northward course *Uttarāyaṇa* and southward course *Dakshiṇāyana*, with 12 *Rāśi* (zodiacal divisions) and *saṁkrānti* moments when the Sun crosses between them. Sāyana (tropical) vs nirayana (sidereal) reference frames are introduced for senior students.

Why this matters

This module is part of an Indian Knowledge Systems curriculum that introduces classical Indian frameworks alongside their modern-scientific parallels. Your child will learn to (a) name and use precise Sanskrit terms, (b) understand the observational basis for the classical framework, and (c) relate it to current scientific or scholarly understanding.

What you can do at home

Safety: never look directly at the Sun. For any sun observation, use projection or proper solar filters. Your child can do sunrise-direction tracking safely from a window.

Conversation prompts

Try one of these over a meal or walk:

1. "*What's one Sanskrit word you learned this week? What does it mean?*"
2. "*What did you find surprising in this module?*"
3. "*Is there anything you're not sure about? Let's look it up together.*"
4. "*How would you explain this to your grandmother / grandfather?*"

Questions parents often ask

Q: Is this religious instruction? A: No. The module presents Indian classical frameworks (cosmology, mathematics, astronomy, ecology) as historical knowledge systems with their own logic and method. The framing is scholarly and comparative, not devotional.

Q: Will my child be tested on Sanskrit? A: Sanskrit terms are introduced for precision, not for rote memorisation. The summative quiz asks students to *use* the terms — for example, “Which doṣa is dominant in the morning hours?” — rather than translate them.

Q: How does this connect to NCERT / board syllabus? A: This module supplements (does not replace) NCERT content. NEP 2020 §4.6 calls for the integration of Indian Knowledge Systems. The pack’s `README.md` lists the specific NEP principles each module advances.

Reading further (for parents)

See `sources.md` in this pack for the full source list.

10-Day Lesson Plan

Lesson Plan — Movement of the Sun (Primary)

10-day arc, 30 min per session.

Day 1 — The Sun’s Yearly Path

Objective: Observe that sunrise direction changes; introduce the ecliptic.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 2 — Axial Tilt and the Seasons

Objective: Demonstrate with torch+globe why we have summer and winter.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 3 — Uttarāyaṇa and Dakshiṇāyana

Objective: Identify the dates and direction of each.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 4 — Equinoxes — Viṣuva

Objective: Identify spring and autumn equinoxes (day = night).

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 5 — Twelve Rāśis

Objective: Name the 12 rāśis in Sanskrit and pair with Greek zodiac names.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 6 — Saṁkrānti Moments

Objective: Identify Makara, Karka, and other saṁkrāntis on the calendar.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 7 — Sāyana vs Nirayana

Objective: Distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 8 — Precession of the Equinoxes

Objective: Explain why Indian and Western zodiac dates differ today.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 9 — Cross-Cultural Solar Calendars

Objective: Compare with Egyptian, Roman, Gregorian.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Day 10 — Final Share and Assessment

Objective: Project share + summative quiz.

Flow: Warm-up (5 min) · Core (15 min) · Activity (8 min) · Wrap-up (5 min)

Pacing notes

- Primary sessions are 30 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see `assessments/project-brief.md`) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see `teacher-notes.md` for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Homework

Homework — Movement of the Sun (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Sun's Yearly Path** · Draw about today's concept. (~5 min)

2. **Day 2 — Axial Tilt and the Seasons** · Draw about today’s concept. (~5 min)
3. **Day 3 — Uttarāyaṇa and Dakshiṇāyana** · Draw about today’s concept. (~5 min)
4. **Day 4 — Equinoxes — Viṣuva** · Draw about today’s concept. (~5 min)
5. **Day 5 — Twelve Rāśis** · Draw about today’s concept. (~5 min)
6. **Day 6 — Saṁkrānti Moments** · Draw about today’s concept. (~5 min)
7. **Day 7 — Sāyana vs Nirayana** · Draw about today’s concept. (~5 min)
8. **Day 8 — Precession of the Equinoxes** · Draw about today’s concept. (~5 min)
9. **Day 9 — Cross-Cultural Solar Calendars** · Draw about today’s concept. (~5 min)
10. **Day 10 — Final Share and Assessment** · Draw about today’s concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.